# Evaluating Batters in Cricket using Markov Chain Models and Survival Analysis

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## Abstract:

Traditional T20 cricket metrics, such as Batting Average and Strike Rate, often fail to capture the situational context essential for modern player evaluation. This research develops a dual-framework approach that combines Markov Chain models with Survival Analysis to evaluate batter performance across various match states. We present a Run Expectation model to assess first-innings maximization and a Win Probability model for second-innings pressure scenarios. Furthermore, a Cox Proportional Hazards model is implemented to quantify Wicket Risk as a dynamic function of match momentum and individual exposure, rather than assuming a static risk of dismissal. Using 16 years of IPL ball-by-ball data, this project offers a high-fidelity alternative to static metrics, providing a dynamic lens through which to view player impact and survival at the crease.

## Introduction:

T20 cricket is characterized by its fast pace and the critical importance of situational context. Despite the sport's growth, player evaluation still relies heavily on conventional metrics like batting average and strike rate. These metrics are limited because they do not account for game states—such as balls remaining, wickets lost, or runs required. For instance, a high strike rate can be misleading if a batter only faces a few deliveries, and a high average may mask slow scoring that harms the team's overall position.

A batter's ability to adapt to these shifting conditions is crucial. This paper introduces a multidimensional modeling approach that uses a Markov Chain framework to calculate Run Expectancy and Win Probability. By defining the game as a finite state space, we can measure a batter's performance relative to the average expected outcome and quantify their ability to transition the team into higher-value states.

## Background:

Cricket, similar to baseball, is well-suited for Markov Chain modeling because it can be analyzed through fixed, sequential states. In this framework, the two innings of a T20 match present unique objectives:

- First Innings: Batters aim to maximize runs within 120 balls, making it ideal for a Run Expectation Model.
- Second Innings: Teams chase a specific target under pressure, requiring a Win Probability Model to evaluate a batter's impact on the likelihood of victory.

While Markov Chain models provide a robust foundation, they typically assume the Markov Property—that the probability of the next event depends only on the current state. However, cricket is highly path-dependent. A batter's risk of dismissal is rarely static and often changes as they become set at the crease.

To address this, we incorporate Survival Analysis using a Cox Proportional Hazards model. Because dismissals are the most volatile events in T20 cricket, causing the largest fluctuations in win probability,

modeling them as a journey is essential. This allows us to predict the instantaneous Hazard Rate (wicket risk) based on situational covariates such as recent dot-ball pressure, team run rate, and match-stage intensity.

## Data:

The analysis in this report is based on ball-by-ball data from the Indian Premier League (IPL) spanning the years 2008 to 2023. Each row in the dataset corresponds to a single ball event and contains various attributes that describe the state of play at that moment. The most important columns used in this analysis include:

*Match.ID*: Unique identifier for each match.

*Over*: The over number in which the ball was bowled.

*Ball*: The specific ball within the over.

*Batter*: The name of the batter facing the delivery.

*Extra.Type*: Indicates any extras (e.g., wide, no-ball) associated with the delivery.

*Innings.Wickets*: The number of wickets lost in the current innings at that point.

*Wicket*: Whether a wicket fell on that particular ball.

*Innings.Runs*: The cumulative runs scored in the innings up to that ball.

*Target.Score*: The target score for the chasing team in the second innings (if applicable).

*Total.Balls.Bowled*: A calculated column that represents the total number of balls bowled in the innings, computed as ((Over - 1) * 6 + Ball).

## Methodology:

### Run Expectation Model and Win Probability Model:

Since both models utilize Markov Chain principles, the methods were very similar and are easily explained in the form of a comparison table.

| Run Expectation Model | Win Probability Model |
|---|---|
| **Data Wrangling** | |
| Only first-inning data was used. | Only second-inning data was used, excluding games decided by the DLS method. |
| **State Space** | |
| The state space included *Total.Balls.Bowled*, *Innings.Wickets, Batter.Runs*, and whether the ball | The state space included *Total.Balls.Bowled*, *Innings.Wickets*, *Total.Runs.Remaining*, and |

| | |
|---|---|
| was an invalid ball (extra). | whether the ball was an invalid ball (extra). |
| **Transition Probabilities** | |
| Transition probabilities between states were calculated by sharing information across states rather than relying solely on empirical probabilities. | *Same procedure as the one applied for the Run Expectation Model* |
| The inning was divided into three stages: balls 1-36 (stage 1), balls 37-84 (stage 2), and balls 85-120 (stage 3). | |
| The probability of each possible event in each stage was calculated and used as transition probabilities to improve representation of rarely visited states. | |
| **Reward Function** | |
| The reward function was equal to the number of *Batter.Runs* in each event. | The reward was 0 for all events except when *Total.Runs.Remaining* in the post state was zero, where it was 1. |
| **Terminal States** | |
| Terminal states were those where *Total.Balls.Bowled* = 120 with no extras, **or** *Innings.Wickets* = 10 | Terminal states were those where *Total.Balls.Bowled* = 120 with no extras, **or** *Innings.Wickets* = 10, **or** *Total.Runs.Remaining = 0* |

The most critical function of a Markov Chain model is the value function. This function calculates the expected long-term cumulative reward from any starting state, up until the process terminates. The value function satisfies the Bellman Equation, which forms the foundation for value iteration. Value iteration, a process proven to converge to the solution of the Bellman Equation, was implemented to calculate the value of each state V(s).

The Bellman Equation is as follows:

$$\textbf{Bellman Equation: } V(s) = \sum_{s'} P(s'|s)[R(s,s') + V(s')]$$

where

- s' is the next possible state from state s, and
- R(s,s') is the reward function for going from state s to s'

For the Run Expectation Model, the value of a state represents the expected number of runs that can be scored from that state until the end of the inning. Conversely, in the Win Probability Model, the value of a state defines the likelihood of the team winning the match given that state.

## Survival Analysis:

To account for the path-dependency and cumulative risk that traditional Markov Models ignore, we employ a Cox Proportional Hazards (CPH) Model. This allows us to treat a batter's innings not just as a series of independent events, but as a survival journey where the risk of dismissal fluctuates based on situational pressure and individual momentum.

*The Hazard Function:*
The primary goal is to model the Hazard Rate, $\lambda(t|X)$, which represents the instantaneous probability of a dismissal at ball $t$, given that the batter has survived up to that point. The model is defined by the formula:

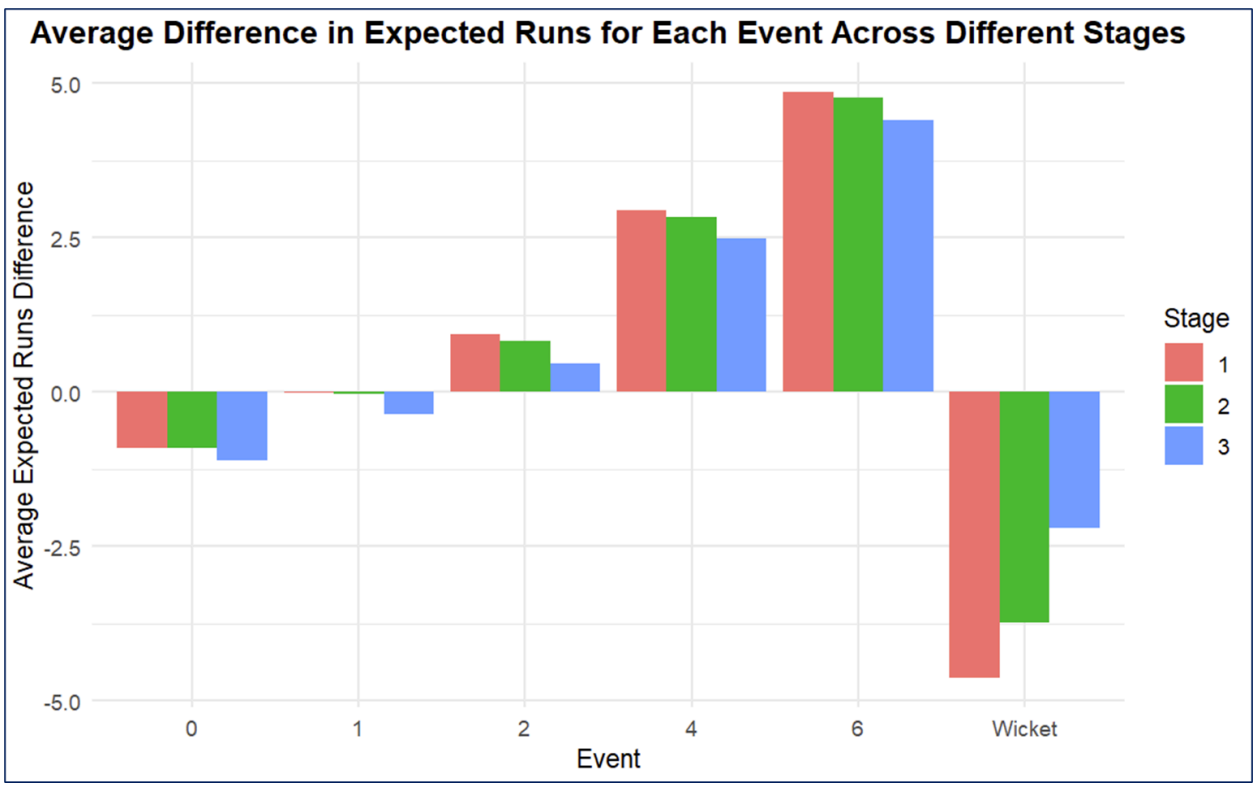
$$\lambda(t|X) = \lambda_0(t)exp(\sum \beta_i X_i)$$

Where:

- $\lambda_0(t|X)$ (Baseline Hazard): The underlying risk of a wicket at time $t$ when all external covariates are at their baseline levels.
- $\exp(\Sigma\beta_i X_i)$ (Hazard Ratio): The relative increase or decrease in risk caused by specific match variables ($X_i$).

## Results:

### Run Expectation Model (1st Innings):




The bar chart illustrates the Average Run Value Added for each possible ball outcome across the three defined stages of the innings: Powerplay (1-36), Middle (37-84), and Death (85-120).

- While boundaries always add value, their impact is highest in Stage 1. A six in the Powerplay provides a greater boost to the final projected total because it sets a higher trajectory for the remaining 14+ overs (84 balls).
- Dot balls consistently yield a negative value added. Interestingly, the penalty for a dot ball is relatively stable across stages, as the consumption of a delivery without scoring always reduces the maximum possible total.
- The model confirms that a wicket in Stage 1 is the most damaging event in the first innings. Losing a top-order batter early doesn't just cost one wicket. It forces a reset in aggression for the remaining duration of the game. Conversely, a wicket in Stage 3 has the smallest negative impact, as there are fewer deliveries remaining for that wicket's absence to be felt.

**Top 10 'Start' Batters by Value Added (VA)**

Minimum 250 balls faced.

| Player | Matches | Balls | Total VA | Avg VA |
|---|---|---|---|---|
| YBK Jaiswal | 21 | 281 | 105.957 | 0.377 |
| SA Yadav | 30 | 346 | 126.069 | 0.364 |
| AB de Villiers | 33 | 290 | 75.023 | 0.259 |
| ML Hayden | 19 | 299 | 72.876 | 0.244 |
| PP Shaw | 32 | 443 | 107.216 | 0.242 |
| CA Lynn | 17 | 260 | 60.913 | 0.234 |
| JC Buttler | 35 | 545 | 124.747 | 0.229 |
| V Sehwag | 47 | 593 | 123.922 | 0.209 |
| RA Tripathi | 29 | 323 | 64.934 | 0.201 |
| F du Plessis | 56 | 731 | 137.618 | 0.188 |

**Top 10 'Middle' Batters by Value Added (VA)**

Minimum 250 balls faced.

| Player | Matches | Balls | Total VA | Avg VA |
|---|---|---|---|---|
| GJ Maxwell | 34 | 344 | 64.367 | 0.187 |
| RR Pant | 23 | 266 | 32.197 | 0.121 |
| SK Raina | 31 | 385 | 40.947 | 0.106 |
| AT Rayudu | 36 | 426 | 43.364 | 0.102 |
| Yuvraj Singh | 43 | 479 | 26.854 | 0.056 |
| DA Miller | 33 | 364 | 10.490 | 0.029 |
| KD Karthik | 44 | 471 | 5.971 | 0.013 |
| AB de Villiers | 40 | 446 | 1.845 | 0.004 |
| RA Jadeja | 38 | 380 | 1.492 | 0.004 |
| YK Pathan | 35 | 366 | −0.863 | −0.002 |

**Top 10 'End' Batters by Value Added (VA)**

Minimum 150 balls faced.

| Player | Matches | Balls | Total VA | Avg VA |
|---|---|---|---|---|
| DJ Bravo | 32 | 198 | 73.060 | 0.369 |
| HH Pandya | 30 | 299 | 103.098 | 0.345 |
| AD Russell | 17 | 185 | 62.635 | 0.339 |
| Harbhajan Singh | 35 | 185 | 41.380 | 0.224 |
| MS Dhoni | 47 | 459 | 90.912 | 0.198 |
| CH Morris | 20 | 169 | 24.292 | 0.144 |
| RA Jadeja | 46 | 370 | 36.748 | 0.099 |
| KD Karthik | 31 | 232 | 18.214 | 0.079 |
| SO Hetmyer | 18 | 200 | 14.278 | 0.071 |
| AR Patel | 31 | 236 | −4.086 | −0.017 |

The above three plots show segmented Run Value Added leaderboards by entry stage. To ensure statistical significance, volume thresholds were applied: a minimum of 250 balls faced for Start and Middle batters, and 150 balls for End batters, accounting for the shorter delivery window in the final stage.

Start Batters (Stage 1 Entry):

- Players like Yashasvi Jaiswal and Suryakumar Yadav dominate the Average Value Added metric. This confirms their reputation as high-intent openers who maximize the Powerplay's potential before the field spreads.

- Notably, legends like Chris Gayle and David Warner appear in the Top 20 but are edged out of the Top 10. This is likely due to their massive volume of balls faced. Over a long career, their Average Value regresses toward the mean, and their higher total count of dismissals (a high-negative value event) pulls down their efficiency compared to the modern sprint style of Jaiswal.

Middle-Order Batters (Stage 2 Entry):

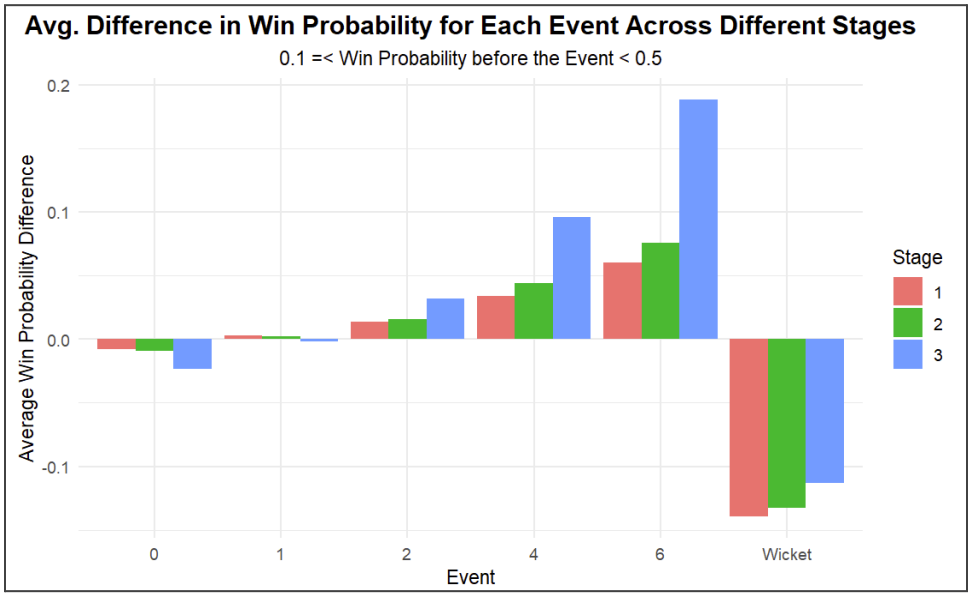
- The results pass the eye test with Glenn Maxwell and Rishabh Pant featuring prominently. Their ability to consistently add positive run value during the middle overs—a period traditionally characterized by consolidation—marks them as elite disruptors.
- The presence of Suresh Raina and Yuvraj Singh underscores the historical value of batters who could transition from strike-rotation to boundary-hitting seamlessly.

End Batters (Stage 3 Entry):

- Dwayne Bravo, Hardik Pandya, and Andre Russell rank highest here. Their value is derived from their ability to convert the final 12–18 balls into scores significantly higher than the baseline expectancy.
- MS Dhoni's inclusion validates the model's ability to capture situational impact. Even with fewer balls to work with, his efficiency in high-expectancy states cements his legacy as a legendary finisher.

An interesting observation from these results is the impact of volume on average value. As seen with Gayle and Warner, facing more deliveries provides more opportunities for negative value events (like wickets or dot balls) to accumulate. To better account for this, future iterations could utilize a Mixed Linear Effect Model. This would allow us to regress individual performance toward the mean based on the total number of balls faced, providing a more shrunken and perhaps more accurate estimate of a player's true underlying talent level regardless of their career length.

### Win Probability Model (2nd Innings):



Unlike the Run Expectation model, where the first stage was most sensitive, the Win Probability model reveals that Stage 3 (the Death overs) is the highest-leverage period for scoring.

- As shown in the bar chart (filtered for high-leverage situations where Win Probability was between 0.1 and 0.5), boundaries in the final stage provide the most significant jumps in win probability. This makes intuitive sense: a six in the 19th over directly negates the pressure of the mounting required run rate.
- Even in a chase, losing a wicket in Stage 1 remains the most detrimental event. An early collapse severely handicaps the team's ability to take the game deep, often dropping the Win Probability significantly before the chase even truly begins.
- The model confirms the hierarchy of value (6 > 4 > 2 > 1), while dot balls act as a constant tax on the team's chances, with the penalty increasing as the ball count dwindles.

**Win Probability Added (WPA): Entry Difficulty = Hard & Entry Stage = Start**

Minimum 300 balls faced.

| Batter | n_balls | n_matches | Total WPA | Avg WPA | Weighted.Entry.Exit.Win.Prob.Diff |
|---|---|---|---|---|---|
| BB McCullum | 317 | 14 | 185.81% | 0.59% | 0.41176978 |
| JC Buttler | 459 | 19 | 256.30% | 0.56% | 0.31337374 |
| AC Gilchrist | 467 | 22 | 252.75% | 0.54% | 0.30351813 |
| N Rana | 399 | 13 | 189.79% | 0.48% | 0.31812885 |
| CH Gayle | 555 | 21 | 254.87% | 0.46% | 0.31296410 |
| DR Smith | 476 | 16 | 196.40% | 0.41% | 0.28932172 |
| RG Sharma | 782 | 33 | 293.69% | 0.38% | 0.22715336 |
| Shubman Gill | 339 | 13 | 124.01% | 0.37% | 0.26131525 |
| AB de Villiers | 408 | 18 | 148.12% | 0.36% | 0.29144045 |
| V Sehwag | 332 | 21 | 119.74% | 0.36% | 0.32069475 |
| KL Rahul | 691 | 19 | 233.46% | 0.34% | 0.25468095 |
| SK Raina | 553 | 28 | 181.49% | 0.33% | 0.37151346 |
| DPMD Jayawardene | 319 | 13 | 99.30% | 0.31% | 0.19380915 |
| CA Lynn | 334 | 13 | 89.86% | 0.27% | 0.25653416 |
| DA Warner | 963 | 38 | 255.99% | 0.27% | 0.22629482 |
| V Kohli | 921 | 29 | 235.78% | 0.26% | 0.28063763 |
| F du Plessis | 602 | 25 | 146.95% | 0.24% | 0.29420635 |
| MA Agarwal | 441 | 22 | 106.46% | 0.24% | 0.23430618 |
| RV Uthappa | 787 | 35 | 170.81% | 0.22% | 0.19375072 |
| G Gambhir | 891 | 34 | 186.98% | 0.21% | 0.36733456 |

To identify the most effective chasers, we analyzed batters who entered during the first 36 balls of the second innings specifically when the team was in a vulnerable position (initial Win Probability between 0.1 and 0.5).

- Brendon McCullum, Jos Buttler, and Adam Gilchrist occupy the top spots. Their high-risk, high-reward approach at the start of a chase is mathematically validated here. By attacking early, they break the back of the chase and skyrocket the team's win probability.
- Similar to the first-innings results, Virat Kohli, Chris Gayle, and David Warner appear in the Top 20. While their average Win Probability added is slightly lower than the sprinters, their sheer volume of successful chases in IPL history is unmatched.

A unique metric introduced in this study is the Average Difference in Win Probability. This measures the state shift a batter creates: where was the team's chance of winning when the batter walked in versus when they walked out?

- McCullum, Raina, and Gautam Gambhir lead this metric. These players are closers in the sense that they rarely leave the team in a worse position than they found it. On average, the win probability of their team is significantly higher upon their dismissal than it was at their arrival, proving their ability to stabilize and then accelerate a chase.

The Win Probability model highlights a crucial tactical truth: while the Death overs are where you win the game (highest value for boundaries), the Powerplay is where you lose it (highest penalty for wickets). The most valuable players in a chase are those who can survive the high-risk early stage while still adding positive value to the Win Probability trajectory.

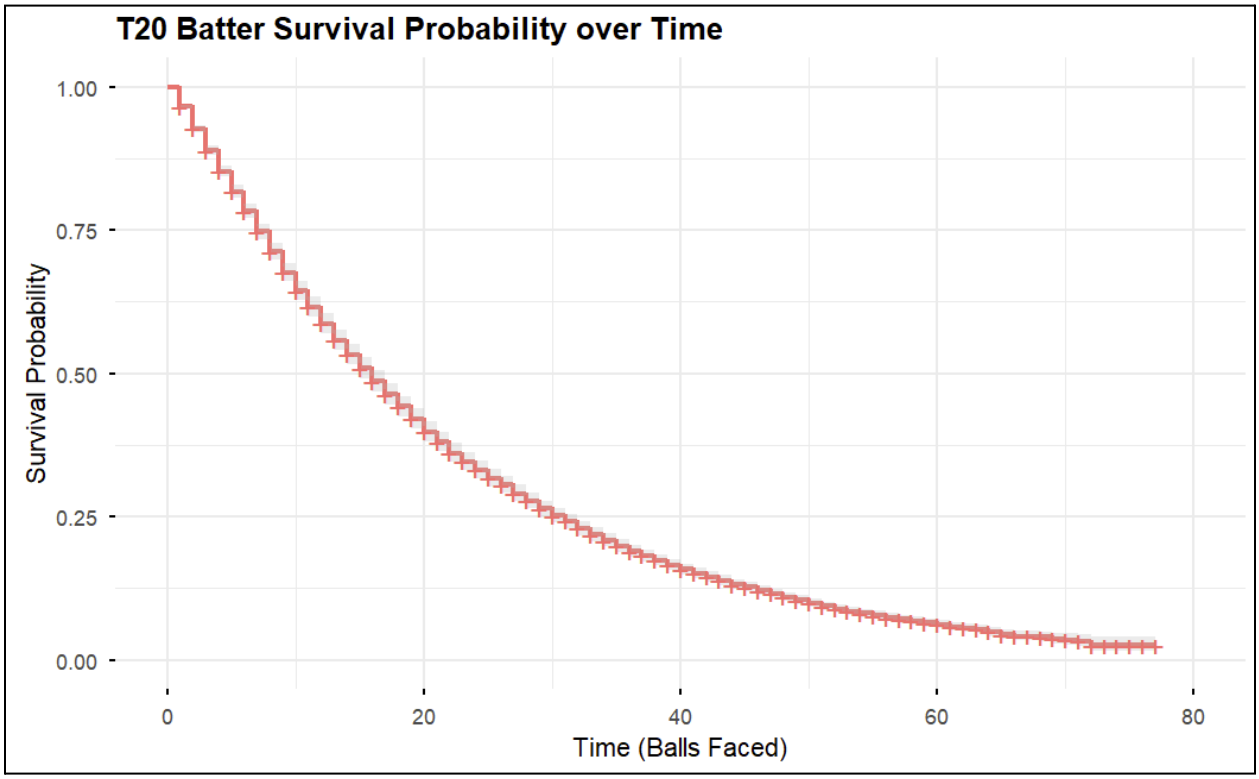
### Cox Proportional Hazards Model:

| | coef | exp(coef) | se(coef) | z | Pr(>\|z\|) | |
|---|---|---|---|---|---|---|
| Innings.Runs.Prev | 0.0046314 | 1.0046422 | 0.0004877 | 9.497 | < 2e-16 | *** |
| Innings.Wickets.Prev | 0.0420076 | 1.0429024 | 0.0065123 | 6.450 | 1.12e-10 | *** |
| Stagemiddle | -0.0713538 | 0.9311324 | 0.0377617 | -1.890 | 0.058814 | . |
| Stageend | 0.3236009 | 1.3820956 | 0.0609349 | 5.311 | 1.09e-07 | *** |
| Batter.SR | 0.0672404 | 1.0695525 | 0.0172961 | 3.888 | 0.000101 | *** |
| Team.RR | -0.1890640 | 0.8277335 | 0.0403334 | -4.688 | 2.77e-06 | *** |
| Total.Batter.Bowler.Balls.Prev | 0.0163612 | 1.0164957 | 0.0042321 | 3.866 | 0.000111 | *** |
| Total.Batter.Dots.L3 | 0.0320974 | 1.0326180 | 0.0127985 | 2.508 | 0.012145 | * |

The model coefficients tell us whether a variable increases or decreases the Wicket Risk. A positive coefficient indicates an increased risk (Hazard Ratio > 1).

- Both Innings.Runs.Prev and Innings.Wickets.Prev have positive coefficients. As the total runs increase, the game moves toward high-intensity stages where risks are necessary. More importantly, as wickets fall, the hazard for the remaining batters increases—likely due to the added pressure of a collapsing lineup and the entry of lower-order players.
- Interestingly, entering in the Middle Stage shows a negative coefficient compared to the Start. This suggests that the new ball period (Stage 1) is inherently more dangerous due to swing and seam movement.

- The risk drastically spikes for Stage 3 entrants (z = 5.311). This is the hit or get out phase where tactical aggression overrides safety.
- A higher individual strike rate correlates with a higher hazard. This mathematically captures the high-risk, high-reward trade-off. The more aggressive the batter, the more likely they are to perish.
- Conversely, if the Team Run Rate is high, the hazard actually decreases. This suggests that when bowlers are leaking runs, the pressure on the batter is lower, or simply that the bowling quality is poor, making survival easier.
- The model shows that as a batter faces more balls from a specific bowler (Batter.Bowler.Balls), the hazard increases. This implies that in T20s, bowlers may adapt to a batter's triggers faster than the batter can solve the bowler.
- One of the most significant findings is that a dot ball in the previous three deliveries increases the hazard. This confirms the psychological impact of scoreboard pressure—dot balls force batters into low-percentage risky shots to break the pressure.




The survival curve illustrates the probability of a batter remaining not out as their innings progresses (balls faced).

- The slope of the survival probability is steeper at the start of the innings. This validates the nervous start hypothesis. Batters are at their most vulnerable during their first few deliveries when they are adjusting to the pace, bounce, and match conditions.

- As the batter faces more balls, the curve flattens. Once a batter is set, their instantaneous risk of dismissal reduces. This is a crucial finding that justifies the Anchor role in T20s—if a batter survives the initial high-hazard period, they become significantly harder to dismiss.

## Future Work:

While this research provides a dynamic framework for evaluating batters, several avenues remain for increasing the precision and predictive power of the models.

- A recurring observation in the player leaderboards was the impact of volume on average value. Prolific batters like David Warner and Chris Gayle faced significantly more deliveries than the current impact leaders, causing their averages to regress toward the mean. Future iterations should employ mixed-effects models to shrink estimates for players with smaller sample sizes toward the population mean. This would provide a more robust talent estimate that accounts for the reliability of the data for each player.
- While the Markov Chain is excellent for state transitions, and the Cox Model excels at hazard prediction, they are still limited by predefined states and linear assumptions. Transitioning to Deep Learning models would allow the system to internalize momentum and pressure as non-linear features. LSTMs are particularly suited for cricket ball-by-ball data, as they can maintain a memory of the entire innings, capturing how a specific sequence of events (e.g., three consecutive boundaries followed by two dots) affects the probability of a wicket in ways a standard Markov transition cannot.
- The current survival model highlights that a batter's risk increases the more they face a specific bowler. Expanding this to include Bowler Type (e.g., Left-arm Pace vs. Leg-spin) and Pitch Conditions would allow for more tactical utility. By integrating bowler-specific hazard ratios, teams could use this model for real-time decision-making, such as identifying the optimal moment to introduce a specific bowler to maximize the Hazard Rate for a set batter.